

Demonstration of Proposed Features

Date	05 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Interactive 3-Step Prediction Wizard	Multi-step guided HTML form with client-side validations to collect applicant details (personal, financial, and loan specifics).	Implemented	Yes	Operates with dynamic progress states and checks for missing inputs.
2	Tuned XGBoost Classifier Predictor	High-accuracy machine learning inference pipeline using a hyperparameter-tuned XGBoost model to evaluate loan eligibility.	Implemented	Yes	Uses pre-trained <code>scale1.pkl</code> and <code>rd1.pkl</code> to compute predictions under 200ms.
3	Relational Database Transaction Logging	System mapping of form inputs, credit scores, and prediction probabilities to SQLite database tables via Flask-SQLAlchemy.	Implemented	Yes	Correctly inserts profiles into relational database tables upon model execution.
4	Cascade-Deletion Audit Logs Management	Administrative screen displaying transaction history with detail views and a secure cascade record deletion trigger.	Implemented	Yes	Successfully removes corresponding user records from SQLite

					tables, avoiding orphan entries.
5	Model Comparison Dashboard	Home screen comparison table comparing training and testing accuracies of Decision Tree, Random Forest, KNN, GB, and XGBoost.	Implemented	Yes	Queries model metadata dynamically from the database to present comparative statistics.
6	Automated Performance Load Testing	Script simulating concurrent virtual user transactions to stress-test routes and monitor response latencies.	Implemented	Yes	Load testing performed using Locust; response latency verified within thresholds.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%